



Robohelp

Mitsubishi tried to sell baby-sized Wakamaru robots, in Japan, in 2003 as domestic helpers, but the product failed

military and healthcare. Typically, in these sectors, tasks that are repetitive, hazardous, or need precision have been taken over by robots. Since the 1980s there's been a growing interest in robots, but it's the 2000s that will be their millennium. So are we going to witness the rise of the robots?

Types of robots

Robots, today, could be autonomous, semi-autonomous or fully controlled, and can broadly be classified into the following types:

1. **Manipulators:** Robotic Manipulators are typically static (fixed-base) robots that are typically used in manufacturing. These are routinely visible in most assembly line factories all over the world, but a sequential phone-switching system could be called a manipulator too.
2. **Mobile robots:** Mobile (movable) robots are usually platforms with



AIBO, the robotic pet was discontinued in 2006

- wheels, tracks or legs. We've already told you about Shakey, and newer examples include Honda's ASIMO (a humanoid that can run at a speed of 6 kmph), and Sony's AIBO (an autonomous pet robot that looks like a dog).
3. **Telerobots:** Tele-operation is when a human operator controls a robot from a distance – of a few feet or halfway across the solar system. The Sojourner robot, which explored the surface of the Mars in 1997, is one such example. Unmanned aerial

Robotics in Digit

Read Dr. Prof. Prahlad Vadakkepat's article on 'Robotics, India and the Future' in the January 2010 issue

vehicles (UAVs) have been around since the Vietnam War, and UGVs (unmanned ground vehicles) and UUVs (unmanned underwater vehicles) have been around since the 1960s, and several are now autonomous. The Mars Rover robots, Spirit and Opportunity, have been working almost independently with timely tele-operated guidance from their NASA mentors since 2003 (only Opportunity remains functional as of now).



Spirit - the NASA Rover for Mars

national event would provide. According to him, this would provide a perfect blend of knowledge and opportunity in addition to an indispensable community that would harness the Indian contribution to the field of robotics. The need of the hour is increased awareness on the activities currently underway in Indian educational communities.

We also spoke to the team behind AcYut, the autonomous robot from BITS Pilani. AcYut won a special prize at the event for registering a World Record in weightlifting – 40 CDs (a standard for weights in robotics events). The team was quite forthcoming in explaining several terminologies associated with robotics. What began as a student-led



(l-r) Samir, Sushma, AcYut and Akash

project has moved across batches of students passing out of BITS Pilani. Newer students are inducted into the team as seniors pass out each year.

Next, we met up with students and professors from countries such as the UK, Singapore and Taiwan to get a feel of what the future holds. One of the people we met was Dr. Tzue-Hseng S. Li, Dean, College of Electrical Engineering and Computer Science, National United University, Taiwan.



Dr. Li pointed at the significance of digital signal processing for logic and AI

Dr. Li played referee for the semi-autonomous and also autonomous Androsot robot soccer events. In the semi-autonomous event, each robot was coded using coloured paper on its

"head". An overhead camera tracks the robot and a server instructs it to move in the required direction. The robot is autonomous – it knows how to kick the ball, how to get up if it trips and falls, and how to chase the ball. The only information provided by

Classification of FIRA robots

The event was segregated into five categories as follows: MiroSot, HuroCup, AndroSot, RoboSot and SimuroSot as defined by FIRA.

MiroSots are rather small robotic bodies with dimensions restricted to 7.5x7.5x7.5cm. These compete in a game of soccer, played with an orange golf ball. The teams are controlled via one host computer per team. Similarly, HuroCup involves physical activities such as sprint, penalty kick, obstacle run, lifting and carrying, weight lifting, marathon, basketball and wall climbing.

AndroSots closely resemble humans to scale and proportion. These are two-legged robots that interpret the position of the ball, goal and members of the opposite team in a game of soccer. Also, these robots are fully autonomous.

RoboSots can be either fully or semi-autonomous. In the semi-autonomous mode, the on-board camera is used by the host computer to guide its movements.